Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Physics-Informed Inverse Estimation of Battery Module Health using Multiscale Digital Twins**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical/Electronics			
	Medium Category	Small Category	Medium Category	Small Category
	Batteries	*Secondary Batteries*		
Concept of Proposed Technology	**Core Concept** The proposed technology focuses on the development of a rapid, non-destructive diagnostic framework for determining the state of health (SOH) of retired electric-vehicle battery modules without requiring full charge–discharge cycling. The core concept is based on the integration of short-duration electrical excitation methods, including sparse electrochemical impedance spectroscopy and controlled current pulse testing, with physics-informed digital twin models augmented by artificial intelligence. These short-time electrical responses contain rich information related to internal electrochemical kinetics, transport limitations, and resistance growth, which can be exploited to infer battery health when appropriately interpreted. **Key Functions** To enable accurate and history-independent diagnosis, reduced-order electrochemical models such as single-particle and pseudo-two-dimensional formulations are employed to generate physics-			

	consistent representations of battery aging across a wide range of degradation states. These models are embedded into artificial intelligence frameworks that invert measured electrical signatures into quantitative health indicators. The diagnostic framework is designed to operate at the battery module level, explicitly accounting for series and parallel cell configurations, electrical interconnections, and internal imbalance effects. As a result, the technology is capable of estimating overall module-level SOH as well as identifying internal health non-uniformity, such as degraded or abnormal cells within a module. **Potential Applications** The proposed technology is intended for deployment in industrial environments such as battery recycling facilities and second-life battery qualification lines, where high throughput, safety, and reliability are essential. By completing diagnostics within five minutes using low-risk electrical excitation, the system enables rapid grading and decision-making for reuse, refurbishment, or recycling of retired EV battery modules.
Rationale for Joint R&D Project	**Critical Technical Barriers** Accurate and fast state-of-health (SOH) assessment of retired electric-vehicle battery modules is a critical bottleneck for battery recycling and second-life deployment. Current industrial practices rely primarily on full charge–discharge testing, which is slow, energy-intensive, and unsuitable for high-throughput screening. Moreover, most retired modules lack reliable usage history, limiting the applicability of conventional data-driven SOH estimation methods and increasing uncertainty in reuse and safety decisions. Addressing this challenge requires a diagnostic framework that can extract health-relevant information from short-duration, non-destructive measurements and validate its performance under realistic, pack-level conditions. The proposed technology demands tight integration of electrochemical physics, reduced-order modeling, artificial intelligence, and diagnostic system engineering. While such methods can be developed at the laboratory scale, their credibility and practical value critically depend on validation using real retired EV battery packs and module-level testing infrastructure. **Why Global Collaboration?** A joint R&D approach is therefore essential to combine advanced modeling and physics-informed AI with access to actual retired EV packs and industrial pack-level testing facilities, enabling rigorous validation and technology maturation. This collaboration allows the diagnostic framework to be evaluated under realistic electrical configurations, aging heterogeneity, and operational constraints, accelerating its transition from laboratory concepts to a TRL-5 prototype demonstrated with real EV battery modules.
Key Words	1. Battery module diagnostics 2. State of health (SOH) estimation 3. Hybrid EIS-pulse testing 4. Physics-informed AI 5. Second-life batteries

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	The objective of this R&D project is to develop a rapid, non-destructive diagnostic technology capable of determining the state of health (SOH) of retired electric-vehicle battery modules within five minutes, without requiring full charge–discharge cycling. The project aims to establish a hybrid diagnostic methodology that combines sparse electrochemical impedance spectroscopy and short-duration current pulse testing with physics-informed digital twin models. Through this approach, the project seeks to achieve accurate and history-independent SOH estimation under safe excitation conditions suitable for industrial environments where throughput, safety, and reliability are critical. In addition to estimating overall module-level SOH, the project aims to identify internal health imbalance and detect degraded or defective cells within a module by leveraging topology-aware inverse modeling and artificial intelligence. The ultimate objective is to demonstrate a fully integrated diagnostic system at Technology Readiness Level 5 using real retired EV battery modules, providing a strong foundation for broader industrial adoption.			
R&D Target Specifications	The target performance specifications of the proposed technology are designed to meet the operational and safety requirements of battery recycling facilities and second-life battery qualification lines. The diagnostic process will be completed in less than 5 minutes per battery module and will not involve full capacity testing or deep cycling. Electrical excitation will be limited to sparse electrochemical impedance measurements combined with one or two short-duration current pulses selected to minimize thermal and safety risks, with the temperature rise during testing maintained below 2 °C. The system is expected to estimate module-level SOH with an error of no more than ±5% compared to reference capacity measurements, while also enabling detection of internal health imbalance or weak cells exhibiting deviations greater than 10% from the module average. The diagnostic framework will be compatible with commercially relevant EV battery modules and will be validated under pilot-scale conditions consistent with TRL-5 requirements.			
R&D Plan	The R&D will be executed through an integrated workflow that connects cell-level aging physics to measurable module-level responses via a hierarchical diagnostic digital twin. This twin combines reduced-order electrochemical models (ROEMs), representing internal degradation mechanisms, with an electrical network model that accounts for the module's series-parallel topology and interconnect resistances. The central strategy is to utilize only the module's main terminals for data acquisition, ensuring the highest possible throughput and deployability in industrial settings. **Phase 1** we will define module-compatible excitation protocols—consisting of sparse Electrochemical Impedance Spectroscopy (EIS) and short current pulses—to excite ohmic, charge-transfer, and diffusion dynamics without significant state-of-charge (SOC) swings. Simultaneously, cell-level degradation modes, such as loss of lithium inventory (LLI) and loss of active material (LAM), will be represented as parameter shifts within the ROEMs. These cell models			

	are embedded into the module's electrical network, which facilitates the transition from cell-level physics to module-level observables. This R&D work treats State of Health (SOH) identification as an inverse problem: estimating the distribution of internal cell health parameters that best explains the aggregated impedance and pulse response measured at the module terminals. This approach allows for the detection of degradation even when it originates at the micro-scale, as the terminal signal encodes a weighted sum of the internal network's state. **Phase 2** we will develop robust inversion algorithms designed to operate under the constraints of terminal-only sensing. The focus will be on the estimation of module SOH and dominant resistance growth signatures with quantified uncertainty, acknowledging that while fine-grained attribution of every individual cell mode is challenging from the terminals, dominant degradation signatures remain identifiable. To support these inversions, physics-based synthetic datasets will be generated over broad aging and temperature ranges to train physics-informed AI surrogate models. These models will rapidly map hybrid EIS–pulse features to SOH and imbalance indicators while providing confidence bounds to ensure conservative decision-making in industrial screening. **Phase 3** The final phase involves integrating this framework into a pilot-scale diagnostic workflow using real retired EV modules. Validation will be performed against reference capacity measurements and selective characterization to verify the accuracy of the SOH estimates. The project will deliver a deployable TRL-5 diagnostic mode that provides fast, terminal-based module SOH assessment and safety flags.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The formulation and validation of electrochemical aging models, the design of hybrid impedance–pulse diagnostic methodologies, and the development of physics-informed digital twin and artificial intelligence frameworks. The institute will conduct simulation-based data generation, laboratory-scale experimental validation, and detailed analysis of degradation mechanisms and diagnostic outputs. In addition, the research institute will contribute to performance benchmarking, interpretation of results, and preparation of technical documentation.
Expected R&R of Korean SME Partner	The Korean SME partner will provide access to real retired EV battery modules and pack-level testing infrastructure required for system-level validation of the proposed framework. Their responsibilities include executing module- and pack-level diagnostic measurements, supplying realistic operational data, and supporting validation against reference performance metrics such as capacity and resistance trends. The industry partner will also contribute to system integration considerations, including measurement constraints, hardware feasibility, and operational workflows, ensuring the developed diagnostic framework is compatible with industrial environments. Through joint analysis with the research institute, partner-provided data will be used to validate, refine, and demonstrate the diagnostic framework at TRL-5 under realistic conditions.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The successful completion of this joint R&D project will result in a novel diagnostic platform that significantly reduces the time, energy, and cost associated with evaluating retired EV battery modules. By enabling accurate SOH estimation within several minutes, the technology will support high-throughput battery grading, improve safety in second-life battery deployment, and enhance decision-making for recycling and reuse pathways. The project is expected to generate patentable technologies related to hybrid diagnostic protocols, physics-informed artificial intelligence algorithms, and module-level health inversion methodologies. More broadly, the outcomes of this project will contribute to strengthening the battery recycling and reuse ecosystem, improving resource efficiency, and supporting sustainable growth in the energy storage industry.